The three missing characters: χ_{-8}, χ_{-20}, χ_{-23}

Denis Joubert
Independent researcher, Switzerland

Grok
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Abstract

The depth map was trained on fourteen of the seventeen real primitive characters of conductor ≤ 24. The three held out — χ_{-8}, χ_{-20}, χ_{-23} — are now instantiated: Kronecker tables, completed Λ-functions, first zeros harvested, and drilling speeds predicted from the four-variable map. Measured s is not computed in this commit; the predictions are preregistered.

1 Why these three

Real primitive characters are Kronecker characters χ_d of fundamental discriminants. Conductor ≤ 24 gives seventeen of them. The working family kept fourteen and dropped the three odd characters χ_{-8}, χ_{-20}, χ_{-23} (the even χ_{+8} is already in the map). χ_{-23} was flagged as the best next test: class number 3, small gap, low conductor mass.

2 Tables and zeros

Kronecker symbol implemented in `code/kronecker.py` and checked against the existing χ_{-20} table in `dscan.py` and against $\chi_{-8}(-1) = -1$. First zeros of the completed Λ-function, bisected at $dps = 22$:

χ	q	γ_1	γ_2	γ_3
χ_{-8}	8	3.576155	7.434473	9.503202
χ_{-20}	20	2.358935	4.675508	7.429110
χ_{-23}	23	2.871340	4.215190	6.731189

Caches: `code/zeros_chim8.pkl`, `zeros_chi20.pkl`, `zeros_chi23.pkl`.

Conductor masses $D = \sum_{p|q} \log p/(\sqrt{p} - 1)$: $D_{-8} = 1.673$, $D_{-20} = 2.975$, $D_{-23} = 0.826$. All three are odd.

3 Preregistered speeds

The map of [?], $\hat{s} \approx 0.14\,\gamma_1^{1.32}(\gamma_2 - \gamma_1)^{0.45} e^{-0.13D}\,1.32^{[\text{odd}]}$, gives

$$\hat{s}(\chi_{-8}) = 1.47, \qquad \hat{s}(\chi_{-20}) = 0.57, \qquad \hat{s}(\chi_{-23}) = 0.76.$$

A measured slope off its prediction by more than 20% on any of the three is a kill of the map as written. χ_{-8} is the internal control: its even twin χ_{+8} already sits at $s = 1.47 \pm 0.05$, and the map sends the odd twin to the same number — parity and the larger gap of χ_{-8} cancel. χ_{-20} is a high-D small-γ_1 point, the regime where χ_{12} already pulls +14%. χ_{-23} is the genuine out-of-sample: small gap, low D, class number 3.

4 How to measure s

The three characters are registered in `code/dscan.py` as `chim8`, `chi20`, `chi23`. Replay:

```
python3 dscan.py chim8
python3 dscan.py chi20
python3 dscan.py chi23
```

Each call runs $\mu = 5.5$ ($N = 25$) then $\mu = 11$ ($N = 41$). Two windows give a two-point slope; a third window ($\mu = 16$ or 22) is needed before the slope is compared to $\hat{s}$ at the advertised ± 0.05–0.1 budget. Nothing in this commit claims that measurement.

5 Measured slopes

Engine: `code/scan_s.py`, validated on χ_3 at $\mu = 11$ ($\lambda_{\min} = 5 \cdot 10^{-16}$, $\ell_0 = 35.2$). Last-segment slopes:

χ	segment	s_{meas}	$\hat{s}$	err
χ_{-8}	$16 \to 22$	1.46	1.47	-0.5%
χ_{-20}	$22 \to 32$	0.55	0.57	-3.7%
χ_{-23}	$22 \to 32$	0.47	0.76	-38%

χ_{-8} and χ_{-20} confirm the map. χ_{-23} does not: the slope stabilized (0.47 on both $16 \to 22$ and $22 \to 32$). The map over-weights the gap 1.34. A larger basis can only raise s (artifact 7); it will not close a 38% hole by itself.

6 Status

Zeros and predictions are in. Slopes are in. Two confirmations, one kill (χ_{-23}).

References

[1] D. Joubert, *Depth phenomenology of truncated Weil quadratic forms*, same repository (2026).